Solo-to-Collaborative Dual-Attention Network for One-Shot Object Detection in Remote Sensing Images

Lingjun Li, Xiwen Yao, *Member, IEEE*, Gong Cheng, *Member, IEEE*, Mingliang Xu, Jungong Han, and Junwei Han

***Abstract*— In this article, we attempt to achieve one-shot object detection by mimicking the human ability to learn new concepts under limited reference, which aims at detecting all object instances of an unseen class in a target image when given a query image of the same unseen class. However, this one-shot learning ability of human benefits from the fact that human brain can quickly extract and process the associated information between the query–target images, which is an issue for the one-shot object detection framework to overcome. Moreover, the feature extraction of the query class in target images is intractable due to the complex and diversified background of remote sensing images. To solve these issues, we propose a solo-to-collaborative dual-attention network (SCoDANet) to hierarchically (image itself/pairs) enhance image feature representations. It consists of three components: 1) solo-attention head that strengthens the compactness of intraclass feature representations of an image and avoids background interference by selectively aggregating the similar features from the spatial and channel dimensions, respectively; 2) dual coattention module that guides RPN to generate an expected set of region proposals related to the query class by mining the coinformation of each query–target feature pair; and 3) nonlinear matching that provides a measure of similarity between the query feature and proposals of the target image to further learn a more robust detector. Our extensive experiments over two benchmarks demonstrate the effectiveness of our method under the one-shot scenario of detecting seen and unseen object categories.**

***Index Terms*— Attention mechanism, object detection, one-shot learning, remote sensing images.**

I. Introduction

OBJECT detection in remote sensing images has achieved great achievements, which can predict the position and category of each object instance in an image [1]–[9]. Remote sensing object detection has a wide range of practical applications, such as geographic information system (GIS) maintenance [10], precision agriculture [11], intelligent monitoring [12], and urban planning [13], [14].

Most of the existing works usually require a large number of labeled samples during the training phase [15]–[17]. Such a large volume of images actually makes it intractable to extend new object categories due to the need for a lot of human annotation labor [18]–[22], whereas it is a truism that humans are capable of learning new concepts with limited guidance. Even if the category of the object is not known in advance, the human visual system can also group the object pixels as a whole according to a given query template, pick up the content clues between the query template and a target image for comparison, and then learn to recognize and locate the objects never seen in the target image, which is dubbed as one-shot learning.

In this article, we expect to construct an object detection model of remote sensing images to realize this learning ability. Therefore, we explore a new task in remote sensing images, which learns to detect an unseen class in a target image with a single query image of the same unseen class by utilizing a two-stage detection framework (e.g., Faster R-CNN [23]), known as one-shot object detection [see Fig. 1(a)]. Here, the unseen class is not included in the training set. In the task of one-shot object detection, the key is to enable the detection model to quickly find out the common category attribute information between the query and target images, just like the human brain, and then integrate these covisual clues to detect all query instances on the target image. However, a regular RPN generates region proposals that are typically over all object classes in a target image, that is, in addition to the query class, there are also other classes in these region proposals. These resulting region proposals are redundant for one-shot object detection. Moreover, the remote sensing images cover a wide area and have complex and diversified backgrounds (see Fig. 1(b), the query class is a car). Even some background objects (such as plants or buildings) are more salient than the query class "car." This makes it tricky to extract the features of the query class from the target image.

We propose a solo-to-collaborative dual-attention network (SCoDANet) to tackle these issues. It can acquire more task-related features of each image and further explore the

Manuscript received April 29, 2021; accepted June 4, 2021. Date of publication July 2, 2021; date of current version January 14, 2022. This work was supported in part by the National Science Foundation of China under Grant 62071388, Grant 61701415, Grant 61772425, and Grant 61773315; and in part the Fundamental Research Funds for the Central Universities and Innovation Foundation for Doctor Dissertation of Northwestern Polytechnical University. *(Corresponding author: Xiwen Yao.)*

Lingjun Li, Xiwen Yao, Gong Cheng, and Junwei Han are with the School of Automation, Northwestern Polytechnical University, Xi'an 710072, China (e-mail: yaoxiwen517@gmail.com).

Mingliang Xu is with the School of Information Engineering, Zhengzhou University, Zhengzhou 450001, China, and also with the Henan Institute of Advanced Technology, Zhengzhou University, Zhengzhou 450001, China (e-mail: iexumingliang@zzu.edu.cn).

Jungong Han is with the Department of Computer Science, Aberystwyth University, Aberystwyth SY23 3FL, U.K. (e-mail: juh22@aber.ac.uk).

Digital Object Identifier 10.1109/TGRS.2021.3091003

1558-0644 © 2021 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 11:17:55 UTC from IEEE Xplore. Restrictions apply.

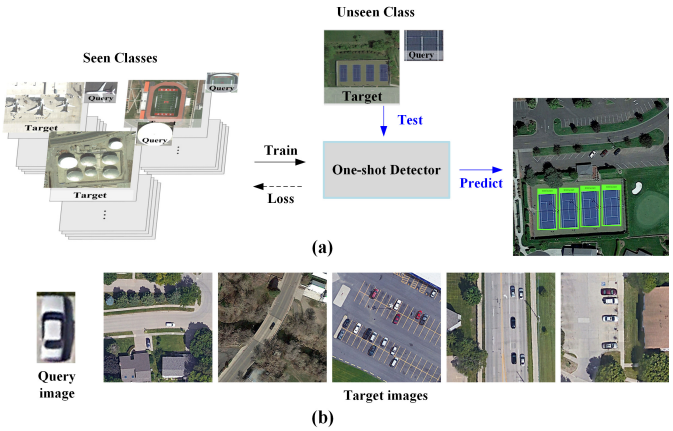


Fig. 1. (a) We train a one-shot object detector on the seen classes and test it on an unseen class (i.e., a tennis court that is unseen during the training phase) with only one labeled sample. (b) When the query class is a car, some different backgrounds in the remote sensing images.

correlation information between each query–target image pair, thereby forming a hierarchical feature enhancement way. Specifically, we first append a solo-attention head on the end of the CNN backbone to comprehensively consider the global dependencies of the target (or query) image features at each position by selectively aggregating associated features in the spatial and channel dimensions, respectively [i.e., spatial solo-attention (SSA) and channel solo-attention (CSA)], thereby strengthening the compactness of intraclass feature representations and avoiding the disturbance of cluttered background. Second, we design a dual coattention (DCoA) module to mine the coinformation among each query–target feature pair by jointly learning the spatialwise and channelwise coattention and enrich the features of interest, thus guiding RPN to yield an expected set of region proposals related to the query class in the target image. DCoA consists of two parallel branches, i.e., a spatial coattention (SCoA) branch and a channel coattention (CCoA) branch. Finally, we formulate a nonlinear matching to obtain a better measure of similarity between the query feature and different proposals of the target image by calculating the absolute difference between the extended query feature and each region proposal feature, and finally learning a more robust detector.

We summarize the contributions of our SCoDANet as follows.

1) We propose an SCoDANet to boost the performance of remote sensing one-shot object detection with a hierarchical feature enhancement manner.
2) A solo-attention head is appended on the end of the backbone to enhance the feature representation of each image. A DCoA module is proposed to explore the associated features between query–target sample pairs, assisting RPN to produce more high-quality and task-related proposals.
3) A nonlinear matching is adapted to further boost one-shot object detection, and SCoDANet achieves leading performance on two benchmarks.

The rest of this article is organized as follows. Section II gives a brief review of remote sensing object detection, one-shot learning, and attention mechanism. In Section III, we describe the proposed SCoDANet in detail. Section IV presents comprehensive experimental results on two benchmarks. Finally, Section V provides the conclusion of this article.

II. Related Works

In this section, we will review related works of our method in three aspects: remote sensing object detection, one-shot learning, and attention mechanism.

A. Remote Sensing Object Detection

Many excellent remote sensing object detection approaches have emerged [24]–[28]. For example, Long *et al.* [29] adapted a region proposal method and CNNs to generate generic image features and then proposed an unsupervised score-based bounding box regression (USB-BBR) method with a nonmaximum suppression to improve location accuracy. Cheng *et al.* [30] proposed a rotation-invariant CNN (RICNN) model to address the issue of object rotation variations by learning a novel rotation-invariant layer and optimizing a specially designed objective function. Li *et al.* [31] proposed an object detection framework that contains RPN and a local-contextual feature fusion network to further solve this problem. Yang *et al.* [32] designed a Markov random field-fully convolutional network (M-FCN) to perform airplane detection in remote sensing images. Xiao *et al.* [33] proposed a novel detection model that includes a GoogleNet, a multiscale fusion feature module, and a support vector machine to detect airports in remote sensing images.

Once these detection models encounter a novel class that is not included in the training set, the detection performance will be greatly degraded. Different from these models, our SCoDANet can improve the generalization ability of the model to novel categories with only one labeled sample.

B. One-Shot Learning

One-shot learning aims at identifying novel object categories by using only one labeled sample. Recently, one-shot learning has been widely used in natural image classification, semantic segmentation, and detection tasks [34]–[38]. Koch *et al.* [39] developed Siamese neural networks for one-shot image classification. Vinyals *et al.* [40] designed matching nets, which is a nonparametric approach or metric learning. Zhang *et al.* [41] proposed a similarity guidance network to obtain the segmentation mask of a query image according to one labeled support image. Hsieh *et al.* [42] developed a coattention and coexcitation (CoAE) model to implement one-shot object detection. All these models mentioned above are applied to the field of natural images, including detection models. Yet, effective models are still to be developed for one-shot object detection in remote sensing images.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 11:17:55 UTC from IEEE Xplore. Restrictions apply.

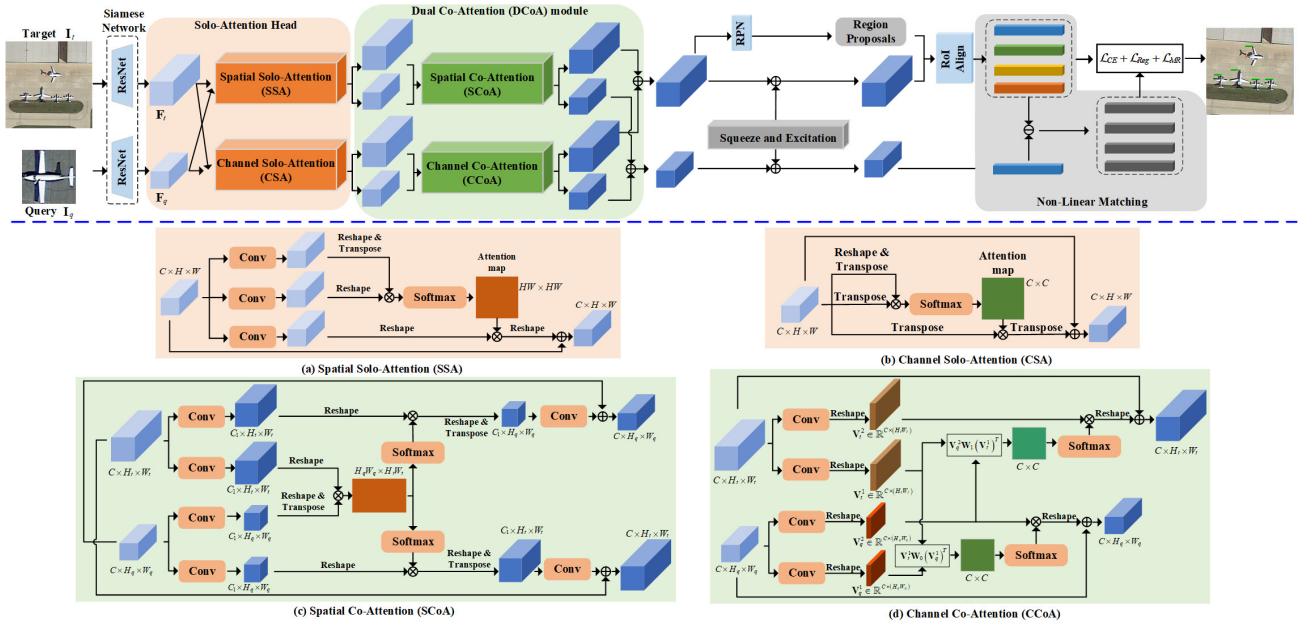


Fig. 2. Overview of our proposed framework, i.e., SCoDANet for one-shot remote sensing object detection. There are three key components: solo-attention head, DCoA network, and nonlinear matching.

C. Attention Mechanism

Attention mechanism has blossomed in the fields of natural language processing and computer vision. For example, Wang *et al.* [43] built nonlocal neural networks to capture long-range dependencies in 3-D video or 2-D images. Du *et al.* [44] improved the nonlocal neural network by adding an interaction-aware loss. Woo *et al.* [45] designed a novel attention module that can generate attention maps along the channel and spatial dimensions. Lu *et al.* [46] developed a new visual question answering (VQA) by using the coattention mechanism to collectively reasoned question and image attention. Similarly, Li *et al.* [47] also introduced the coattention mechanism into VQA. Li *et al.* [48] proposed a coattention recurrent unit to produce groupwise attention among all images and performed object cosegmentation. Lu *et al.* [49] designed a coattention Siamese network for the unsupervised video object segmentation. We introduce the coattention mechanism into our SCoDANet to explore the associated features between query–target sample pairs in one-shot remote sensing object detection.

III. Proposed Method

A. Overview of the Proposed Method

In this work, we propose an SCoDANet that can learn better image feature representation and explore coattention between query–target sample pairs, guiding RPN to yield a region proposal set related to the query class on the target sample feature. As illustrated in Fig. 2, for an input query–target sample pair, we exploit ResNet [50] as a backbone to extract the features of the query and target images and then feed these features into two parallel attention branches. First, the SSA head processes the query and target features, respectively, to enrich contextual information about their respective local features. Then, these processed features are fed into the SCoA branch, which can build spatialwise associations among query–target feature pairs to explore more cofeatures' information. Second, the query and target features also input into the CSA head to improve the feature representation of class-specific semantics by exploring interdependencies among channels. Then, we feed these processed features into the CCoA branch to mine the channelwise correlations between the query and target features. We merge these coattention features with the raw query–target features in an elementwise sum manner for generating appropriate region proposals. Finally, we utilize a nonlinear matching to further improve the detection performance.

In this work, there are two kinds of different datasets, i.e., a training set D_{seen} and a testing set D_{unseen}. Their label sets, respectively, are L_{seen} and L_{unseen}, and nonintersect each other ($L_{\text{seen}} \cap L_{\text{unseen}} = \emptyset$). In the testing phase, a query sample $\mathbf{I}_q$ of a unseen class is given, and this unseen class l_q is not included in the training set. We detect all instances of this unseen class l_q in a target image $\mathbf{I}_t$.

B. Solo-Attention Head

Considering the complex and varied background of remote sensing images, we append a solo-attention head on the end of the CNN backbone to enhance the compactness of intraclass feature representations and weaken the disturbance of cluttered background. The solo-attention head mainly includes an SSA head and a CSA head [51]. Given a query feature $\mathbf{F}_q = \Phi(\mathbf{I}_q) \in \mathbb{R}^{C\times H_q \times W_q}$ and a target feature $\mathbf{F}_t = \Phi(\mathbf{I}_t) \in \mathbb{R}^{C\times H_t \times W_t}$, Φ represents the conventional feature extraction network.

1) Spatial Solo-Attention Head: For the query feature $\mathbf{F}_q$, we compute an attention matrix $\mathbf{M}_0^q$ of $\mathbf{F}_q$ by

$$\mathbf{M}_0^q = \left(f_0(\mathbf{F}_q, \theta_0)\right)^T \times g_0(\mathbf{F}_q, \varphi_0) \in \mathbb{R}^{(H_q W_q)\times(H_q W_q)}. \quad (1)$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 11:17:55 UTC from IEEE Xplore. Restrictions apply.

$f_0(\cdot)$ and $g_0(\cdot)$ indicate two embedding functions that are implemented as 1×1 convolution kernel. θ_0 and φ_0 are learnable parameters. Then, inputting $\mathbf{M}_0^q$ into a softmax layer, we can obtain an attention map $\mathbf{S}_0^q \in \mathbb{R}^{(H_qW_q)\times(H_qW_q)}$

$$s_{ij} = \frac{\exp\left(\mathbf{M}_0^q\right)_{ij}}{\sum_{j=1}^{H_qW_q}\left(\mathbf{M}_0^q\right)_{ij}} \tag{2}$$

where s_{ij} is an element of the ith row and the jth column of $\mathbf{S}_0^q$. Meanwhile, the query feature $\mathbf{F}_q$ is fed into a convolution layer to yield a new feature, which can be multiplied by $\mathbf{S}_0^q$. Next, we integrate the resulting feature with $\mathbf{F}_q$ to get the final output $\bar{\mathbf{F}}_{q_ss} \in \mathbb{R}^{C\times H_q\times W_q}$ as follows:

$$\bar{\mathbf{F}}_{q_ss} = \beta\left(h\left(\mathbf{F}_q,\alpha\right)\times\left(\mathbf{S}_0^q\right)^T\right) + \mathbf{F}_q. \tag{3}$$

$h(\cdot)$ is similar to $f_0(\cdot)$, and α is its parameters. β is a learnable parameter. Likewise, for the target feature $\mathbf{F}_t$, we perform the above procedure to get the final target feature $\bar{\mathbf{F}}_{t_ss} \in \mathbb{R}^{C\times H_t\times W_t}$.

2) Channel Solo-Attention Head: Different from SSA, CSA strengthens the channel information in the input features. For the query feature $\mathbf{F}_q$, instead of adapting the convolution layer, we directly use the original feature $\mathbf{F}_q$ to calculate the attention matrix $\mathbf{M}_1^q$ of $\mathbf{F}_q$

$$\mathbf{M}_1^q = \text{reshape}\left(\mathbf{F}_q\right)\times\left(\text{reshape}\left(\mathbf{F}_q\right)\right)^T \in \mathbb{R}^{C\times C} \tag{4}$$

where reshape($\mathbf{F}_q$) changes the size of $\mathbf{F}_q$ to $C \times (H_qW_q)$. Next, we input $\mathbf{M}_1^q$ into a softmax layer to obtain an attention map $\mathbf{S}_1^q \in \mathbb{R}^{C\times C}$

$$s_{ij} = \frac{\exp\left(\mathbf{M}_1^q\right)_{ij}}{\sum_{j=1}^{C}\left(\mathbf{M}_1^q\right)_{ij}} \tag{5}$$

where s_{ij} is an element of the ith row and the jth column of $\mathbf{S}_1^q$. Then, $\mathbf{S}_1^q$ be multiplied by $\mathbf{F}_q$, and we perform an elementwise sum operation with $\mathbf{F}_q$ to produce the final output $\bar{\mathbf{F}}_{q_cs} \in \mathbb{R}^{C\times H_q\times W_q}$ by

$$\bar{\mathbf{F}}_{q_cs} = \lambda\left(\mathbf{S}_1^q \times \text{reshape}\left(\mathbf{F}_q\right)\right) + \mathbf{F}_q. \tag{6}$$

Analogously, for the target feature $\mathbf{F}_t$, we perform the above procedure to get the final target feature $\bar{\mathbf{F}}_{t_cs} \in \mathbb{R}^{C\times H_t\times W_t}$.

C. Dual Coattention

We design a DCoA module by introducing the coattention mechanism to mine the coattention among each query–target sample pair, guiding RPN to generate more region proposals related to the query class on the target sample feature. As shown in Fig. 2, DCoA contains an SCoA branch and a CCoA branch, which can jointly learn the spatialwise and channelwise coattention, thus better exploring the associated features between query–target sample pair. Note that, through the solo-attention head, we can obtain the enhanced query features ($\bar{\mathbf{F}}_{q_ss}$ and $\bar{\mathbf{F}}_{q_cs}$) and target features ($\bar{\mathbf{F}}_{t_ss}$ and $\bar{\mathbf{F}}_{t_cs}$).

1) Spatial Coattention Branch: We compute the spatialwise coattention matrix $\mathbf{A}_s$ between $\bar{\mathbf{F}}_{q_ss}$ and $\bar{\mathbf{F}}_{t_ss}$

$$\mathbf{A}_s = f_1\left(\bar{\mathbf{F}}_{t_ss},\theta_1\right)\times\left(g_1\left(\bar{\mathbf{F}}_{q_ss},\varphi_1\right)\right)^T \in \mathbb{R}^{\left(H_qW_q\right)\times(H_tW_t)} \tag{7}$$

where $f_1(\cdot)$ and $g_1(\cdot)$ are similar to $f_0(\cdot)$, and θ_1 and φ_1 are their parameters, respectively. $f_1(\bar{\mathbf{F}}_{t_ss},\theta_1) \in \mathbb{R}^{C_1\times(H_tW_t)}$ and $g_1(\bar{\mathbf{F}}_{q_ss},\varphi_1) \in \mathbb{R}^{C_1\times(H_qW_q)}$ are flattened into matrix representations, and C_1 is channel dimension, $C_1 < C$, for reducing computation amount. Then, this matrix $\mathbf{A}_s$ is combined with the query feature $\bar{\mathbf{F}}_{q_ss}$ and target feature $\bar{\mathbf{F}}_{t_ss}$, respectively, to obtain the corresponding spatialwise coattention features $\mathbf{F}'_{q_SCo}$ and $\mathbf{F}'_{t_SCo}$, which is formulated as

$$\begin{aligned}\mathbf{F}'_{q_SCo} &= \phi\left(\left(\frac{1}{H_tW_t}\mathbf{A}_s\right)\times f\left(\bar{\mathbf{F}}_{t_ss},\theta\right),\omega_0\right) \in \mathbb{R}^{C\times H_q\times W_q}\\ \mathbf{F}'_{t_SCo} &= \psi\left(\left(\frac{1}{H_qW_q}\mathbf{A}_s^T\right)\times f\left(\bar{\mathbf{F}}_{q_ss},\theta\right),\omega_1\right) \in \mathbb{R}^{C\times H_t\times W_t}\end{aligned} \tag{8}$$

where $\phi(\cdot)$ and $\psi(\cdot)$ denote two embedding functions with batch normalization, which are implemented as 1×1 convolution kernel. ω_0 and ω_1 are learnable parameters. $f(\cdot)$ with θ as its parameter is similar to $f_0(\cdot)$. Next, $\bar{\mathbf{F}}_{q_ss}$ ($\bar{\mathbf{F}}_{t_ss}$) and $\mathbf{F}'_{q_SCo}$ ($\mathbf{F}'_{t_SCo}$) are integrated together to form the corresponding new extended features

$$\begin{aligned}\hat{\mathbf{F}}_{q_SCo} &= \bar{\mathbf{F}}_{q_ss} + \mathbf{F}'_{q_SCo}\\ \hat{\mathbf{F}}_{t_SCo} &= \bar{\mathbf{F}}_{t_ss} + \mathbf{F}'_{t_SCo}\end{aligned} \tag{9}$$

2) Channel Coattention Branch: Different from the SCoA branch, this branch focuses on the channelwise coattention information. Similarly, we feed a query feature $\bar{\mathbf{F}}_{q_cs}$ and a target feature $\bar{\mathbf{F}}_{t_cs}$ into the CCoA branch. Let $\bar{\mathbf{F}}_{t_cs}$ be the input reference; the channelwise coattention matrix $\mathbf{A}_c^0$ related to $\bar{\mathbf{F}}_{q_cs}$ can be obtained by

$$\mathbf{A}_c^0 = \left(f_2\left(\bar{\mathbf{F}}_{t_cs},\theta_2\right)\times\mathbf{W}_0\right)\times\left(g_2\left(\bar{\mathbf{F}}_{q_cs},\varphi_2\right)\right)^T \in \mathbb{R}^{C\times C} \tag{10}$$

where $f_2(\cdot)$ and $g_2(\cdot)$ are similar to $f_0(\cdot)$, and θ_2 and φ_2 are their parameters, respectively. $\mathbf{W}_0$ is a learnable weight matrix and implemented by using a fully connected layer with $(H_tW_t) \times (H_qW_q)$ parameters. Then, combine $\mathbf{A}_c^0$ with the query feature $\bar{\mathbf{F}}_{q_cs}$ to obtain the channelwise coattention feature $\mathbf{F}'_{q_CCo}$ by

$$\mathbf{F}'_{q_CCo} = \left(\frac{1}{C}\mathbf{A}_c^0\right)\times g\left(\bar{\mathbf{F}}_{q_cs},\varphi\right) \in \mathbb{R}^{C\times H_q\times W_q} \tag{11}$$

where $g(\cdot)$ with φ as its parameter is also an embedding function and similar to $g_0(\cdot)$. Analogously, let $\bar{\mathbf{F}}_{q_cs}$ be the input reference; the channelwise coattention matrix $\mathbf{A}_c^1$ related to $\bar{\mathbf{F}}_{t_cs}$ can be obtained by

$$\mathbf{A}_c^1 = \left(f_3\left(\bar{\mathbf{F}}_{t_cs},\theta_3\right)\right)^T\times\left(g_3\left(\bar{\mathbf{F}}_{q_cs},\varphi_3\right)\times\mathbf{W}_1\right) \in \mathbb{R}^{C\times C} \tag{12}$$

where $\mathbf{W}_1 \in \mathbb{R}^{(H_qW_q)\times(H_tW_t)}$ is consistent with $\mathbf{W}_0$ in the implementation manner. Then, combine $\mathbf{A}_c^1$ with the target feature $\bar{\mathbf{F}}_{t_cs}$ to obtain the channelwise coattention feature $\mathbf{F}'_{t_CCo}$

$$\mathbf{F}'_{t_CCo} = \left(\frac{1}{C}\mathbf{A}_c^1\right)\times f_4\left(\bar{\mathbf{F}}_{t_cs},\theta_4\right) \in \mathbb{R}^{C\times H_t\times W_t} \tag{13}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 11:17:55 UTC from IEEE Xplore. Restrictions apply.

TABLE I

EVALUATION OF UNSEEN CLASS SET 1 AND CLASS SET 2 ON DIOR IN AP (%). "BB FIELD" REPRESENTS "BASEBALL FIELD." "BKB COURT" REPRESENTS "BASKETBALL COURT." "ES AREA" REPRESENTS "EXPRESSWAY SERVICE AREA." "ET STATION" REPRESENTS "EXPRESSWAY TOLL STATION." "GT FIELD" REPRESENTS "GROUND TRACK FIELD"

Method	Class Set 1	ResNet50			ResNet101		Class Set 2	ResNet50		
		OneshotDet [54]	CoAE [42]	Ours	CoAE [42]	Ours		OneshotDet [54]	CoAE [42]	Ours
Seen Class	airplane	46.2	52.98	52.90	51.29	53.86	airport	42.5	78.33	82.31
	airport	42.6	75.87	76.37	77.91	77.57	BKB court	51.2	80.62	80.17
	BB field	64.6	66.02	65.57	66.22	67.00	bridge	26.5	41.34	40.89
	BKB court	43.7	80.72	80.69	80.94	80.53	dam	43.6	62.30	63.46
	bridge	26.0	38.69	40.34	42.45	42.45	ES area	56.9	68.23	68.76
	ES area	58.3	68.76	68.12	71.18	76.45	ET station	37.8	56.93	57.28
	ET station	39.7	57.18	55.39	57.38	59.16	harbor	33.9	37.35	37.49
	golf field	58.3	75.01	77.38	78.33	78.16	golf field	59.5	76.36	76.53
	GT field	51.4	77.45	79.92	80.22	80.28	overpass	47.5	57.13	57.51
	overpass	46.6	56.04	58.01	59.11	58.68	ship	60.9	40.95	45.77
	stadium	54.5	67.18	76.00	73.86	75.01	stadium	47.0	66.49	71.29
	storage tank	51.9	43.47	42.84	43.79	43.78	storage tank	52.4	43.58	43.56
	tennis court	68.4	71.49	77.35	80.10	79.72	train station	36.2	61.52	62.14
	train station	34.7	59.93	60.44	64.02	63.68	vehicle	29.1	32.32	32.50
	vehicle	26.4	31.87	32.14	32.44	33.29	windmill	74.7	80.03	80.32
	mAP	47.55	61.51	**62.90**	63.95	**64.64**	mAP	46.65	58.90	**59.99**
Unseen Class	chimney	3.2	4.04	4.44	4.68	7.84	airplane	3.0	8.53	6.67
	dam	17.6	20.57	27.26	18.83	22.43	BB field	2.9	6.51	15.06
	harbor	2.3	1.44	1.99	1.57	1.77	chimney	3.0	8.08	5.81
	ship	1.7	8.64	8.50	7.51	8.71	GT field	8.5	17.22	21.00
	windmill	0.7	5.76	7.26	6.96	5.69	tennis court	1.2	0.17	0.26
	mAP	5.1	8.09	**9.89**	7.91	**9.29**	mAP	3.72	8.10	**9.76**
mAP in total		36.94	48.15	**49.65**	49.94	**50.80**	mAP in total	35.92	46.20	**47.44**

where $f_4(\cdot)$ is similar to $f_0(\cdot)$ and θ_4 is its parameter. We integrate $\bar{\mathbf{F}}_{q_cs}$ ($\bar{\mathbf{F}}_{t_cs}$) and $\mathbf{F}'_{q_CCo}$ ($\mathbf{F}'_{t_CCo}$) together to form the corresponding new extended features

$$\begin{aligned}\hat{\mathbf{F}}_{q_CCo} &= \bar{\mathbf{F}}_{q_cs} + \lambda_q \mathbf{F}'_{q_CCo} \\ \hat{\mathbf{F}}_{t_CCo} &= \bar{\mathbf{F}}_{t_cs} + \lambda_t \mathbf{F}'_{t_CCo}\end{aligned} \tag{14}$$

where λ_q and λ_t are learnable parameters.

Finally, we combine the outputs of the SCoA and CCoA branches to form the final feature representations with coattention

$$\begin{aligned}\tilde{\mathbf{F}}_q &= \hat{\mathbf{F}}_{q_SCo} + \hat{\mathbf{F}}_{q_CCo} \\ \tilde{\mathbf{F}}_t &= \hat{\mathbf{F}}_{t_SCo} + \hat{\mathbf{F}}_{t_CCo}.\end{aligned} \tag{15}$$

$\tilde{\mathbf{F}}_t$ can be fed into RPN to generate more region proposals closed to the query sample. In $\tilde{\mathbf{F}}_q$, the pivotal features of $\mathbf{F}_q$ are enhanced, and its background noise is weakened. Hence, leveraging $\tilde{\mathbf{F}}_q$ can improve classification accuracy.

D. Query-Proposal Feature Learning

Through the DCoA module, we obtain the extended query feature $\tilde{\mathbf{F}}_q \in \mathbb{R}^{C\times H_q \times W_q}$ and the target feature $\tilde{\mathbf{F}}_t \in \mathbb{R}^{C\times H_t \times W_t}$; these features contain coattention information. In consideration of the same channel dimension, the correlation between $\tilde{\mathbf{F}}_q$ and $\tilde{\mathbf{F}}_t$ can be further explored by exploiting the squeeze-and-excitation (SE) technique [52], which is formulated as

$$\mathrm{SE}\big(\tilde{\mathbf{F}}_q\big) = \mathbf{w} \in \mathbb{R}^C, \tilde{\mathbf{F}}'_t = \mathbf{w} \odot \tilde{\mathbf{F}}_t,\ \tilde{\mathbf{F}}'_q = \mathbf{w} \odot \tilde{\mathbf{F}}_q \tag{16}$$

where $\odot$ is an elementwise product. In the Region-of-Interest (RoI) pooling phase, we map the candidate proposals generated by RPN to the resulting target feature $\tilde{\mathbf{F}}'_t$ to get the fixed-size proposal regions. Then, these proposal regions and the query feature $\tilde{\mathbf{F}}'_q$ are, respectively, fed into the convolution module

$$\begin{aligned}\mathbf{r} &= \mathrm{convs}\big(\mathrm{roipool}\big(\mathbf{proposals}, \tilde{\mathbf{F}}'_t\big)\big) \in \mathbb{R}^{N\times C_2} \\ \mathbf{q} &= \mathrm{convs}\big(\tilde{\mathbf{F}}'_q\big) \in \mathbb{R}^{1\times C_2}\end{aligned} \tag{17}$$

where $\mathbf{r}$ denotes the proposal features and $\mathbf{q}$ is the processed query feature. N is the number of proposals. Next, we perform a nonlinear matching that calculates the absolute difference between the query feature $\mathbf{q}$ and each region proposal feature $r_i \in \mathbf{r}$ $(i = 1, \ldots, N)$ and then inputs this difference and each region proposal feature into the classifier to calculate the classification score

$$\begin{aligned}\mathbf{X} &= \big[|\mathrm{expand}(\mathbf{q}) - \mathbf{r}|, \mathbf{r}\big] \in \mathbb{R}^{N\times 2C_2} \\ \mathrm{Scores} &= \mathrm{Classifier}(\mathbf{X})\end{aligned} \tag{18}$$

where $\mathrm{expand}(\mathbf{q}) \in \mathbb{R}^{N\times C_2}$.

In addition to the cross-entropy and regression losses of Faster R-CNN, we also introduce a margin-based ranking loss proposed by [42]

$$\begin{aligned}\mathcal{L}_{\mathrm{MR}}(\{x_i\}) &= \sum_{i=1}^{K} \left\{ \begin{matrix} y_i \times \max\{m^+ - s_i, 0\} \\ +(1-y_i) \times \max\{s_i - m^-, 0\} + \Delta_i \end{matrix} \right\} \\ \Delta_i &= \sum_{j=i+1}^{K} \left\{ \begin{matrix} [y_i = y_j] \times \max\{|s_i - s_j| - m^-, 0\} \\ +[y_i \neq y_j] \times \max\{m^+ - |s_i - s_j|, 0\} \end{matrix} \right\}\end{aligned} \tag{19}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 11:17:55 UTC from IEEE Xplore. Restrictions apply.

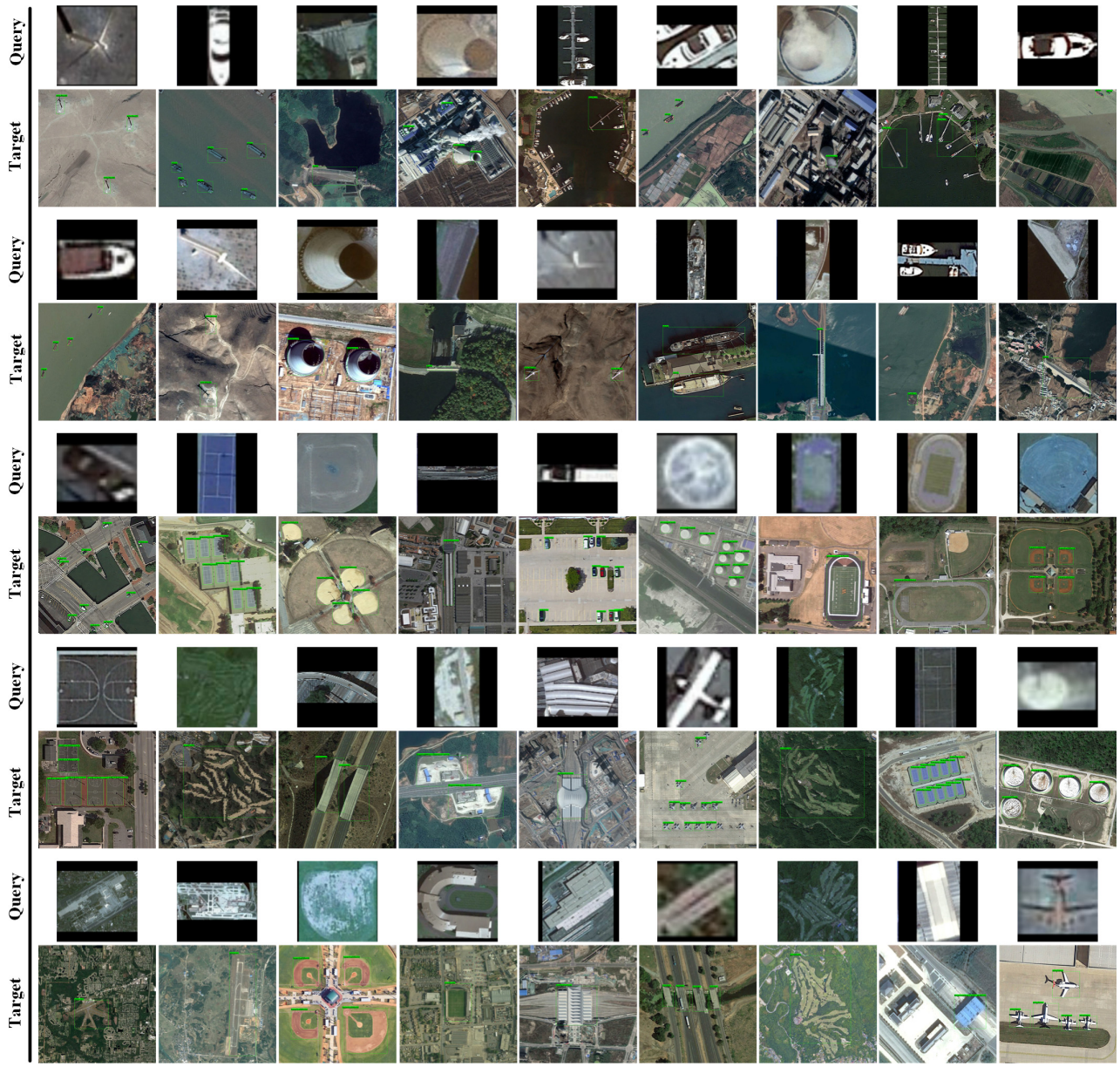


Fig. 3. Some test sample results of unseen class set 1 on the DIOR dataset. The first two "Query" rows indicate unseen classes; the corresponding "Target" rows are detection results. The remaining "Query" rows indicate seen classes.

where $x_i \in \mathbf{X}$. $K = 128$ indicates the number of the possible candidate proposals chosen by RPN. $s = \text{Classifier}(\mathbf{X})$ is the foreground probability in regard to the query $\mathbf{q}$. m^+ and m^- are the upper and lower limits of the expected probabilities, respectively. Therefore, the training procedure is accomplished by optimizing a joint objective function

$$\arg\min\left\{\mathcal{L}_{\text{CE}} + \mathcal{L}_{\text{Reg}} + \delta\mathcal{L}_{\text{MR}}\right\} \tag{20}$$

where $\mathcal{L}_{\text{CE}}$ denotes the cross-entropy loss and $\mathcal{L}_{\text{Reg}}$ is the regression loss. δ is the regularization factor. Note that we train our model on the seen classes, and then, we transfer the learned model parameters to detect both the seen classes and unseen classes in the test phase.

IV. Experiments

In this section, we perform contrast experiments and ablation studies to verify the effectiveness of our SCoDANet.

A. Datasets

DIOR has 23 463 images and 190 288 instances, including 20 object classes, which is a large-scale, publicly available database for object detection in optical remote sensing images [15]. In this database, all images are collected from Google Earth (Google Inc.), and the spatial resolutions range from 0.5 to 30 m. The following information is the classes contained in the dataset: airplane, airport, baseball field, basketball court, bridge, chimney, dam, expressway service area, expressway toll station, harbor, golf course, ground track field, overpass, ship, stadium, storage tank, tennis court, train station, vehicle, and windmill. For the DIOR dataset, we resize the size of all images to 800 × 800.

NWPU VHR-10.v2 is released by Northwestern Polytechnical University in 2014 [53]. The data include 1172 images, covering ten object categories. The object categories include

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 11:17:55 UTC from IEEE Xplore. Restrictions apply.

TABLE II

EVALUATION OF UNSEEN CLASS SET 1 AND UNSEEN CLASS SET 2 ON NWPU VHR-10.V2 IN AP (%). "GT FIELD" REPRESENTS "GROUND TRACK FIELD"

Method	Class Set 1	ResNet50			ResNet101		Class Set 2	ResNet50		
		OneshotDet [54]	CoAE [42]	Ours	CoAE [42]	Ours		OneshotDet [54]	CoAE [42]	Ours
Seen Class	basketball	41.3	75.18	79.68	77.13	78.49	airplane	96.0	99.25	99.56
	bridge	53.1	79.23	86.18	74.70	76.50	baseball	90.9	90.91	90.89
	GT field	97.0	100.00	99.84	100.00	100.00	bridge	67.6	79.28	76.05
	harbor	74.8	94.54	87.07	79.85	95.42	harbor	77.4	89.51	88.10
	ship	90.4	87.00	90.04	89.18	89.70	ship	86.7	84.40	85.67
	storage tank	96.9	88.24	90.81	85.49	90.83	storage tank	95.4	87.42	90.85
	vehicle	56.0	77.86	77.66	78.14	78.86	tennis court	78.9	67.63	78.93
	mAP	72.79	86.01	**87.32**	83.50	87.11	mAP	84.7	85.48	**87.15**
Unseen Class	airplane	7.3	9.12	18.50	2.76	9.08	basketball	14.2	3.35	3.09
	baseball	2.9	19.47	16.81	0.86	9.72	GT field	0.00	0.07	1.14
	tennis court	9.5	4.10	6.75	12.41	7.04	vehicle	7.6	26.31	35.24
	mAP	6.57	10.89	**14.02**	5.34	8.61	mAP	7.27	9.91	**13.16**
mAP	in total	52.92	63.47	**65.33**	60.05	63.56	mAP in total	61.47	62.81	**64.95**

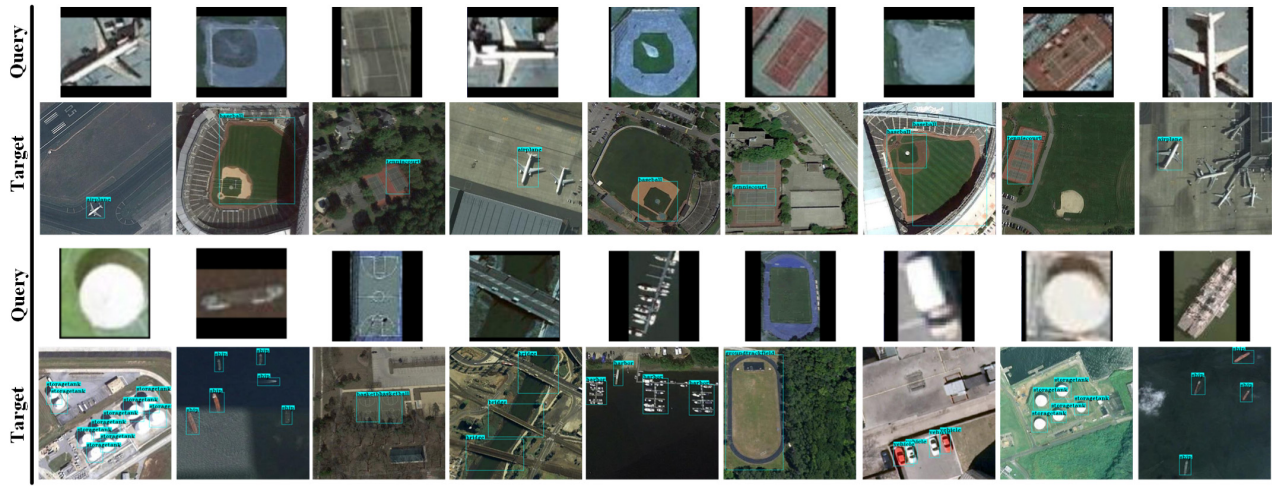


Fig. 4. Some test sample results of unseen class set 1 on the NWPU-VHR10.v2 dataset. The first two rows indicate unseen classes and detection results. The last two rows indicate seen classes.

airplane, baseball diamond, basketball, bridge, court, ground track field, harbor, ship, storage tank, tennis court, and vehicle. In our experiments, all images of this dataset are resized to 400 × 400.

B. Implementation Details

SCoDANet is implemented by using the Pytorch framework. We adapt the pretrained ResNet-50 (or ResNet-101) provided in [42] as a backbone to extract the query and target image features. To ensure that the proposed detection model does not "see" the unseen classes in the training stage, we delete the images containing both the seen and unseen classes in the data preprocessing stage. We embed SCoDANet into Faster R-CNN. SGD is used to optimize our detection model, and the momentum parameter is 0.9. We set the learning rate to 0.01. We train our SCoDANet for 40 epochs. For (19), we set $m^+ = 0.7$ and $m^- = 0.3$ in all experiments. Following Hsieh *et al.* [42], we crop out the ground-truth bounding boxes of the class-specific objects as the query samples that are resized to 128 × 128 in all experiments, among which only the area of the ground-truth bounding box is larger than 400 pixels, are selected. In the testing phase, we repeat the test four times and take the average value as the final detection result.

C. Experimental Results and Comparisons

In our experiments, we choose CoAE [42] and OneshotDet [54] as the baseline methods and carry out experiments on different datasets. In each dataset, we evaluate our SCoDANet on different splits of unseen classes.

1) Evaluation on DIOR: We randomly selected five categories as unseen classes for testing. The remaining 15 classes

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 11:17:55 UTC from IEEE Xplore. Restrictions apply.

TABLE III
ABLATION STUDY OF UNSEEN CLASS SET 1 USING RESNET50 ON DIFFERENT DATASETS IN AP (%)

SCoA	CCoA	SSA	CSA	Non-Linear Matching	DIOR		NWPU VHR-10.v2	
					mAP of unseen classes	mAP of all classes	mAP of unseen classes	mAP of all classes
√				√	9.56	49.09	12.71	64.58
√		√			10.32	49.56	6.56	61.9
√		√		√	10.51	49.91	13.68	63.84
	√				8.72	48.07	7.40	61.83
	√			√	9.80	48.58	10.27	63.96
	√		√		8.84	47.80	9.10	61.92
	√		√	√	10.43	48.91	15.03	64.20
√	√				9.23	49.44	6.92	62.19
√	√			√	9.95	49.39	7.81	61.92
√	√	√			10.72	49.48	5.77	61.92
√	√	√		√	9.39	49.50	10.53	63.47
√	√		√		8.77	48.60	4.91	61.44
√	√		√	√	9.82	49.34	12.60	64.10
√	√	√	√		10.53	49.28	7.46	62.75

are used as seen classes for training. In Table I, we first perform one-shot object detection on the unseen class set 1 on the DIOR dataset. When we adopt a pretrained network ResNet50 as the backbone, compared with CoAE and OneshotDet, our SCoDANet yields the highest result of 49.65% mAP on all classes. For unseen classes (or seen classes), the detection performance of SCoDANet is still superior to CoAE and OneshotDet. Furthermore, we also use a pretrained ResNet101 as backbone; our DCoASNet beats CoAE by 0.86% on all 20 classes. More specifically, our DCoASNet obtains 9.29% mAP in unseen classes and exceeds the baseline CoAE by 1.38%. For seen classes, DCoASNet achieves 64.64% mAP, which outperforms CoAE by a margin. Moreover, we also evaluate our model on an unseen class set 2 (see Table I). As can be observed, our SCoDANet consistently outperforms CoAE and OneshotDet by a large margin on all seen classes and unseen classes. It is also obvious that our SCoDANet would not sacrifice the mAP of seen classes in one-shot object detection. Some detection results of unseen class set 1 on the DIOR dataset are shown in Fig. 3. From these detection results, we can observe that our method can yield relatively accurate detection boxes for the unseen classes. However, we can also find that there is a gap between the performance of the unseen classes and that of the seen classes. We believe that the performance of the unseen classes may be further improved by solving the issues of rotational diversity and scale variability of objects in the remote sensing image.

2) Evaluation on NWPU VHR-10.v2: In the unseen class set 1, we randomly select three classes as unseen classes. The remaining seven classes are considered as seen classes. From Table II, by adapting ResNet50 as the backbone, OneshotDet and CoAE, respectively, yield 52.92% and 63.47% in all classes, whereas our SCoDANet boosts the performance of all classes to 65.33%. Note that, the detection performance gain of unseen classes is obtained by our method compared to CoAE (14.02% versus 10.89%). Likewise, our SCoDANet gets the detection performance of seen classes 87.32%; OneshotDet and CoAE obtain 72.79% and 86.01%. When we use a pretrained ResNet101 as the backbone, our DCoASNet obtains 63.56% mAP in all classes and surpasses CoAE by 3.51%. For unseen classes, our DCoASNet yields 8.61%, reaping a significant performance gain (approximately 3.27% mAP) against CoAE. Besides, we also perform experiments on an unseen class set 2. The experimental results are shown in Table II. Compared with OneshotDet and CoAE, our method achieves significant performance improvement. In Fig. 4, we visualize some detection results of unseen class set 1 on the NWPU VHR-10.v2 dataset.

D. Ablation Analysis

To verify the performance of each component in SCoDANet, we implement ablation experiments in Table III. In Table III, "SCoA" means the SCoA branch. "CCoA" indicates the CCoA branch. "SSA" stands for the spatial solo-attention head. "CSA" represents the channel solo-attention head. "Nonlinear matching" appertains to the calculation of the absolute difference between each region proposal feature and the extended query feature. On the DIOR dataset, compared with CoAE that obtains 48.15% mAP for all classes and 8.09% mAP for unseen classes, employing "SCoA" and "nonlinear matching" yields 9.56% mAP for the unseen classes and 49.09% mAP for all classes, which brings 1.47% and 0.94% improvements. Adding the "SSA" head for "SCoA" can precede the baseline CoAE by 2.23% mAP for the unseen classes and 1.41% mAP for all classes. When we integrate "SCoA," "SSA" and "nonlinear matching" together, the performance further improves to 10.51% mAP for the unseen classes and 49.91% mAP for all classes. Employing "CCoA," individually outperforms the baseline CoAE by 0.63% mAP for the unseen classes. Adding "nonlinear matching" on this basis can beat the baseline CoAE by a large margin. Analogously, integrating "CCoA," "CSA," and "nonlinear matching"

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 11:17:55 UTC from IEEE Xplore. Restrictions apply.

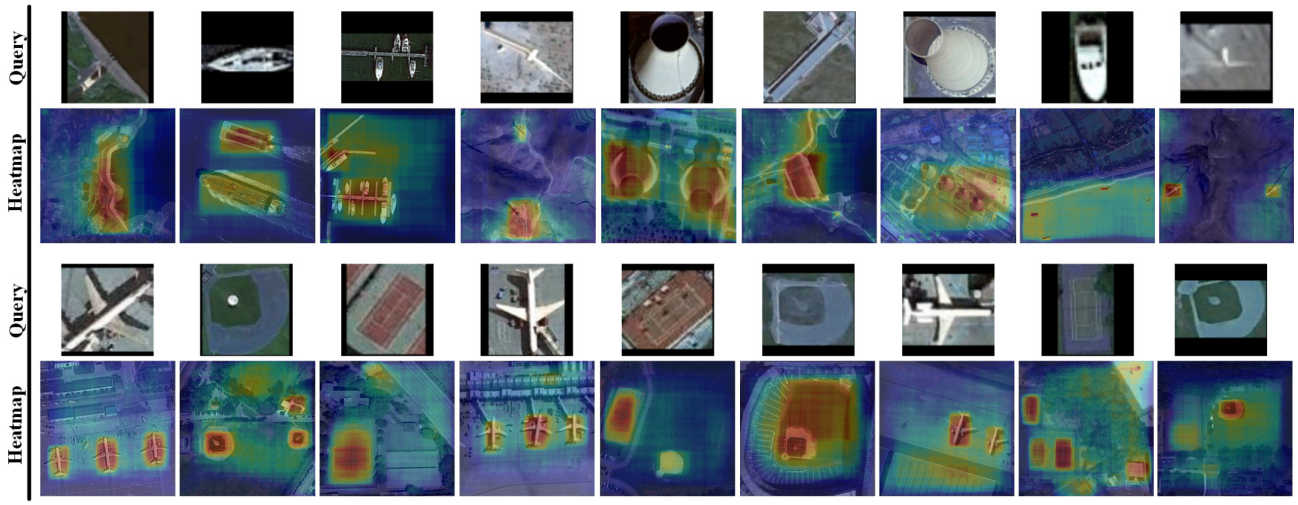


Fig. 5. Distribution of object proposals of unseen classes. The first two rows indicate unseen classes from the unseen class set 1 on the DIOR dataset and heatmaps. The last two rows indicate unseen classes from the unseen class set 1 on the NWPU-VHR10.v2 dataset and heatmaps.

together can achieve performance improvement. In addition, we also employ "SCoA" and "CCoA" jointly significantly outperforms the baseline CoAE. Then, we insert a "nonlinear matching" head on this basis, and the detection performance of the unseen classes has been further improved. We add the "SSA" head and the "CSA" head for "SCoA" and "CCoA" successively and obtain better detection performance than the baseline CoAE. We also perform ablation experiments on the NWPU VHR-10.v2 dataset, and the experimental results are shown in Table III. By observing these results, we find that using any module designed by us on the DIOR dataset would improve the detection performance of the unseen classes, and integrating these modules would further improve the detection performance of the seen classes and unseen classes.

The distribution of object proposals of the unseen classes generated by RPN that is guided by SCoDANet can be visualized in Fig. 5. Follow Hsieh *et al.* [42], we generate distribution heatmaps of object proposals by counting the number of region proposals that cover each pixel and normalizing these counts to a probability map. From Fig. 5, we can see that employing SCoDANet can enable the RPN to focus on the regions related to the query class, thus generating more high-quality region proposals in the target image.

In addition, we carry out a comparative experiment of computational efficiency on DIOR and NWPU VHR-10.v2, with ResNet50 as a backbone. *First,* Table IV *shows the processing time per iteration on the training phase. From* Table IV, *we can see that the training processing time of our method is slightly longer than that of the baselines, but we can train a high-accuracy, one-shot object detector that significantly outperforms the baselines. In addition, inference speed comparison is shown in* Table V. As shown in Table V, although our method adds an extra dual-attention module, the FPS reduction is very small. For example, on the class set 1 (or class set 2) of DIOR, compared with CoAE, the computational efficiency of our method dropped from 9.43 (or 9.10) to 8.76 (or 9.04) frames per second (FPS). The same is true for the inference speed on NWPU VHR-10.v2. To sum up, with a small increase in the computational amount, the proposed dual-attention module can effectively acquire and process the coinformation between the query and target images, and achieve the tradeoff between precision and complexity.

TABLE IV
TRAINING TIME PER ITERATION ON DIOR AND NWPU VHR-10.v2

Method	*DIOR (s)*		*NWPU VHR-10.v2 (s)*	
	Class Set 1	*Class Set 2*	*Class Set 1*	*Class Set 2*
CoAE[42]	*0.6406*	*1.2552*	*0.8738*	*0.8462*
OneshotDet[54]	*1.0178*	*1.0074*	*0.4691*	*0.6874*
Ours	*1.2880*	*0.8153*	*1.0627*	*1.1324*

TABLE V
INFERENCE SPEED ON DIOR AND NWPU VHR-10.v2

Method	DIOR (Speed/fps)		NWPU VHR-10.v2 (Speed/fps)	
	Class Set 1	Class Set 2	Class Set 1	Class Set 2
CoAE[42]	9.43	9.10	19.67	19.82
OneshotDet[54]	5.89	5.91	3.76	3.45
Ours	8.76	9.04	18.44	18.60

V. CONCLUSION

In this work, we propose an SCoDANet for one-shot remote sensing object detection. Specifically, we insert a solo-attention head on the end of the backbone to better represent image features. Then, we also design a DCoA module that assists RPN to generate more appropriate region proposals. Finally, we implement a nonlinear matching operation to learn a more robust detector. Experimental results on the DIOR

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 11:17:55 UTC from IEEE Xplore. Restrictions apply.

and NWPU VHR-10.v2 datasets demonstrate that SCoDANet can achieve excellent detection performance on unseen and seen classes. In addition, how to extract the rotation invariant and scale-invariant feature representation is a challenge for one-shot object detection in remote sensing images. In our future work, we intend to design a rotation-scale equivariant network to extract more robust feature representations, thus bridging the gap between the performance of the unseen classes and that of the seen classes.

REFERENCES

[1] Y. Gong *et al.*, "Context-aware convolutional neural network for object detection in VHR remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 34–44, Jan. 2020.

[2] J. Han, X. Yao, G. Cheng, X. Feng, and D. Xu, "P-CNN: Part-based convolutional neural networks for fine-grained visual categorization," *IEEE Trans. Pattern. Anal. Mach. Intell.*, early access, Aug. 6, 2019, doi: 10.1109/TPAMI.2019.2933510.

[3] X. Yao, J. Han, D. Zhang, and F. Nie, "Revisiting co-saliency detection: A novel approach based on two-stage multi-view spectral rotation co-clustering," *IEEE Trans. Image Process.*, vol. 26, no. 7, pp. 3196–3209, Jul. 2017.

[4] Z. Sun, H. Fang, M. Deng, A. Chen, P. Yue, and L. Di, "Regular shape similarity index: A novel index for accurate extraction of regular objects from remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 7, pp. 3737–3748, Jul. 2015.

[5] C. Chen, B. Zhang, A. Del Bue, and V. Murino, "Manifold constrained low-rank decomposition," in *Proc. IEEE Int. Conf. Comput. Vis. Workshops (ICCVW)*, Oct. 2017, pp. 1800–1808.

[6] Y. Shen *et al.*, "Efficient deep learning of non-local features for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, early access, Aug. 17, 2020, doi: 10.1109/TGRS.2020.3014286.

[7] C. Chen, W. Li, E. W. Tramel, and J. E. Fowler, "Reconstruction of hyperspectral imagery from random projections using multihypothesis prediction," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 1, pp. 365–374, Jan. 2014.

[8] Y. Zhong, X. Han, and L. Zhang, "Multi-class geospatial object detection based on a position-sensitive balancing framework for high spatial resolution remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 138, pp. 281–294, Apr. 2018.

[9] X. Han, Y. Zhong, and L. Zhang, "An efficient and robust integrated geospatial object detection framework for high spatial resolution remote sensing imagery," *Remote Sens.*, vol. 9, no. 7, p. 666, Jun. 2017.

[10] Y. Liu, J. Yao, X. Lu, M. Xia, X. Wang, and Y. Liu, "RoadNet: Learning to comprehensively analyze road networks in complex urban scenes from high-resolution remotely sensed images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2043–2056, Apr. 2019.

[11] X. Yao, J. Han, G. Cheng, X. Qian, and L. Guo, "Semantic annotation of high-resolution satellite images via weakly supervised learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 6, pp. 3660–3671, Jun. 2016.

[12] Q. Li, L. Mou, Q. Liu, Y. Wang, and X. X. Zhu, "HSF-Net: Multiscale deep feature embedding for ship detection in optical remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 12, pp. 7147–7161, Dec. 2018.

[13] P. Zhong and R. Wang, "A multiple conditional random fields ensemble model for urban area detection in remote sensing optical images," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 12, pp. 3978–3988, Dec. 2007.

[14] S. Shi, Y. Zhong, J. Zhao, P. Lv, Y. Liu, and L. Zhang, "Land-use/land-Cover change detection based on class-prior object-oriented conditional random field framework for high spatial resolution remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, early access, Nov. 16, 2020, doi: 10.1109/TGRS.2020.3034373.

[15] K. Li, G. Wan, G. Cheng, L. Meng, and J. Han, "Object detection in optical remote sensing images: A survey and a new benchmark," *ISPRS J. Photogramm. Remote Sens.*, vol. 159, pp. 296–307, Jan. 2020.

[16] J. Pang, C. Li, J. Shi, Z. Xu, and H. Feng, "R^2-CNN: Fast tiny object detection in large-scale remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5512–5524, Aug. 2019.

[17] P. Lv, Y. Zhong, J. Zhao, and L. Zhang, "Unsupervised change detection based on hybrid conditional random field model for high spatial resolution remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 7, pp. 4002–4015, Jul. 2018.

[18] X. Feng, J. Han, X. Yao, and G. Cheng, "TCANet: Triple context-aware network for weakly supervised object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, early access, Oct. 26, 2020, doi: 10.1109/TGRS.2020.3030990.

[19] X. Feng, J. Han, X. Yao, and G. Cheng, "Progressive contextual instance refinement for weakly supervised object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 11, pp. 8002–8012, Nov. 2020.

[20] X. Yao, X. Feng, J. Han, G. Cheng, and L. Guo, "Automatic weakly supervised object detection from high spatial resolution remote sensing images via dynamic curriculum learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 675–685, Jan. 2021.

[21] K. Ronald, R. Luu, and C. Kanan, "Low-shot learning for the semantic segmentation of remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 10, pp. 6214–6223, Oct. 2018.

[22] G. Cheng, C. Yang, X. Yao, L. Guo, and J. Han, "When deep learning meets metric learning: Remote sensing image scene classification via learning discriminative CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 5, pp. 2811–2821, May 2018.

[23] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, Jun. 2017.

[24] C. Zhou, J. Zhang, J. Liu, C. Zhang, G. Shi, and J. Hu, "Bayesian transfer learning for object detection in optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 11, pp. 7705–7719, Nov. 2020.

[25] X. Peng, R. Zhong, Z. Li, and Q. Li, "Optical remote sensing image change detection based on attention mechanism and image difference," *IEEE Trans. Geosci. Remote Sens.*, early access, Nov. 10, 2020, doi: 10.1109/TGRS.2020.3033009.

[26] Y. Yu, X. Yang, J. Li, and X. Gao, "A cascade rotated anchor-aided detector for ship detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, early access, Dec. 7, 2020, doi: 10.1109/TGRS.2020.3040273.

[27] Y. Hu, J. Chen, D. Pan, and Z. Hao, "Edge-guided image object detection in multiscale segmentation for high-resolution remotely sensed imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 8, pp. 4702–4711, Aug. 2016.

[28] H. He, Y. Lin, F. Chen, H.-M. Tai, and Z. Yin, "Inshore ship detection in remote sensing images via weighted pose voting," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 6, pp. 3091–3107, Jun. 2017.

[29] Y. Long, Y. Gong, Z. Xiao, and Q. Liu, "Accurate object localization in remote sensing images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 5, pp. 2486–2498, May 2017.

[30] G. Cheng, P. Zhou, and J. Han, "Learning rotation-invariant convolutional neural networks for object detection in VHR optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 12, pp. 7405–7415, Dec. 2016.

[31] K. Li, G. Cheng, S. Bu, and X. You, "Rotation-insensitive and context-augmented object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 4, pp. 2337–2348, Apr. 2018.

[32] Y. Yang, Y. Zhuang, F. Bi, H. Shi, and Y. Xie, "M-FCN: Effective fully convolutional network-based airplane detection framework," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 8, pp. 1293–1297, Aug. 2017.

[33] Z. Xiao, Y. Gong, Y. Long, D. Li, X. Wang, and H. Liu, "Airport detection based on a multiscale fusion feature for optical remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 9, pp. 1469–1473, Sep. 2017.

[34] A. Osokin, D. Sumin, and V. Lomakin, "OS2D: One-stage one-shot object detection by matching anchor features," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Aug. 2020, pp. 635–652.

[35] L. Fei-Fei, R. Fergus, and P. Perona, "One-shot learning of object categories," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 28, no. 4, pp. 594–611, Apr. 2006.

[36] L. Fe-Fei and Perona, "A Bayesian approach to unsupervised one-shot learning of object categories," in *Proc. 9th IEEE Int. Conf. Comput. Vis.*, Oct. 2003, pp. 1134–1141.

[37] S. Caelles, K.-K. Maninis, J. Pont-Tuset, L. Leal-Taixe, D. Cremers, and L. Van Gool, "One-shot video object segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 5320–5329.

[38] L. Li, J. Han, X. Yao, G. Cheng, and L. Guo, "DLA-MatchNet for few-shot remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, early access, Nov. 6, 2020, doi: 10.1109/TGRS.2020.3033336.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 11:17:55 UTC from IEEE Xplore. Restrictions apply.

[39] G. Koch, R. Zemel, and R. Salakhutdinov, "Siamese neural networks for one-shot image recognition," in *Proc. Int. Conf. Mach. Learn. (ICML) Deep Learn. Workshop*, Jul. 2015, pp. 1–8.
[40] O. Vinyals, C. Blundell, T. Lillicrap, K. Kavukcuoglu, and D. Wierstra, "Matching networks for one shot learning," in *Proc. Neural Inf. Process. Syst. (NeurIPS)*, 2016, pp. 3630–3638.
[41] X. Zhang, Y. Wei, Y. Yang, and T. S. Huang, "SG-one: Similarity guidance network for one-shot semantic segmentation," *IEEE Trans. Cybern.*, vol. 50, no. 9, pp. 3855–3865, Sep. 2020.
[42] T.-I. Hsieh, Y.-C. Lo, H.-T. Chen, and T.-L. Liu, "One-shot object detection with co-attention and co-excitation," in *Proc. Neural Inf. Process. Syst. (NeurIPS)*, 2019, pp. 2725–2734.
[43] X. Wang, R. Girshick, A. Gupta, and K. He, "Non-local neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7794–7803.
[44] Y. Du, C. Yuan, B. Li, L. Zhao, Y. Li, and W. Hu, "Interaction-aware spatio-temporal pyramid attention networks for action classification," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 373–389.
[45] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 3–19.
[46] J. Lu, J. Yang, D. Batra, and D. Parikh, "Hierarchical question-image co-attention for visual question answering," in *Proc. Neural Inf. Process. Syst. (NeurIPS)*, 2016, pp. 289–297.
[47] X. Li *et al.*, "Beyond RNNs positional self-attention with co-attention for video question answering," in *Proc. Assoc. Advancement Artif. Intell. (AAAI)*, 2019, pp. 8658–8665.
[48] B. Li, Z. Sun, Q. Li, Y. Wu, and H. Anqi, "Group-wise deep object co-segmentation with co-attention recurrent neural network," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 8519–8528.
[49] X. Lu, W. Wang, C. Ma, J. Shen, L. Shao, and F. Porikli, "See more, know more: Unsupervised video object segmentation with co-attention Siamese networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3623–3632.
[50] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
[51] A. Vaswani *et al.*, "Attention is all you need," in *Proc. Neural Inf. Process. Syst. (NeurIPS)*, 2017, pp. 6000–6010.
[52] J. Hu, L. Shen, S. Albanie, G. Sun, and E. Wu, "Squeeze-and-excitation networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 42, no. 8, pp. 2011–2023, Aug. 2020.
[53] G. Cheng, J. Han, P. Zhou, and L. Guo, "Multi-class geospatial object detection and geographic image classification based on collection of part detectors," *ISPRS J. Photogramm. Remote Sens.*, vol. 98, pp. 119–132, Dec. 2014.
[54] X. Li, L. Zhang, Y. Pun Chen, Y.-W. Tai, and C.-K. Tang, "One-shot object detection without fine-tuning," 2020, *arXiv:2005.03819*. [Online]. Available: http://arxiv.org/abs/2005.03819



Lingjun Li received the M.S. degree from Shaanxi Normal University, Xi'an, Shaanxi, China, in 2018. She is pursuing the Ph.D. degree with Northwestern Polytechnical University, Xi'an.

Her research interests include computer vision and remote sensing image processing, especially on object detection and scene classification.



Xiwen Yao (Member, IEEE) received the B.S. and Ph.D. degrees from Northwestern Polytechnical University, Xi'an, China, in 2010 and 2016, respectively.

He is an Associate Professor with Northwestern Polytechnical University. His research interests include computer vision and remote sensing image processing, especially on fine-grained image classification and object detection.



Gong Cheng (Member, IEEE) received the B.S. degree from Xidian University, Xi'an, China, in 2007, and the M.S. and Ph.D. degrees from Northwestern Polytechnical University, Xi'an, in 2010 and 2013, respectively.

He is a Professor with Northwestern Polytechnical University. His main research interests are computer vision and pattern recognition.



Mingliang Xu received the B.S. and M.S. degrees from Zhengzhou University, Zhengzhou, China, in 2005 and 2008, respectively, and the Ph.D. degree from Zhejiang University, Hangzhou, China, in 2012.

He is a Full Professor with Zhengzhou University. His research interests include computer graphics, multimedia, and artificial intelligence.



Jungong Han is a Full Professor and the Chair in Computer Science with Aberystwyth University, Aberystwyth, U.K.

He has published over 180 articles, including more than 40 IEEE TRANSACTIONS and more than 40 A* conference papers. His research interests span the fields of video analysis, computer vision, and applied machine learning.



Junwei Han received the B.S., M.S., and Ph.D. degrees in pattern recognition and intelligent systems from Northwestern Polytechnical University, Xi'an, China, in 1999, 2001, and 2003, respectively.

He is a Professor with Northwestern Polytechnical University. His research interests include computer vision and brain-imaging analysis.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 11:17:55 UTC from IEEE Xplore. Restrictions apply.